

Getting started hints - for students

This section provides a few brief hints for the student in how to begin thinking about analysing the data.

The main approach is through regression, with DO as the response variable and a variety of potential explanatory variables. A key part of the data structure is the `id` variable. When the boat goes out to take measurements it does so at all the sampling stations on the same day, so those measurements with the same `id` are all subject to whatever special weather and water conditions may be at work that day. So `id` defines a random effect which should be included in the model.

It will be helpful to begin by subsetting the data to include only a single depth. The depth closest to the surface is the natural one to start with. Models can be fitted to lower depths if time permits.

It is likely that linear regression will not fully capture the patterns in the data. This is a complex river system so we should expect non-linear changes with time and space. Additive models which allow smooth terms in the regression model are likely to be helpful. These models are available in the `mgcv` package for R. A good introduction is provided in the book *Generalized Additive Models : An Introduction with R*, Second Edition, by Simon N. Wood, published by CRC press.